MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, STRAIGHT, SELF-LOCKING AND NONSELF-LOCKING, CATEGORY 4B (FOR MIL-C-38999 SERIES I AND II CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the accessories described herein shall consist of this specification sheet and the latest issue of MIL-C-85049.

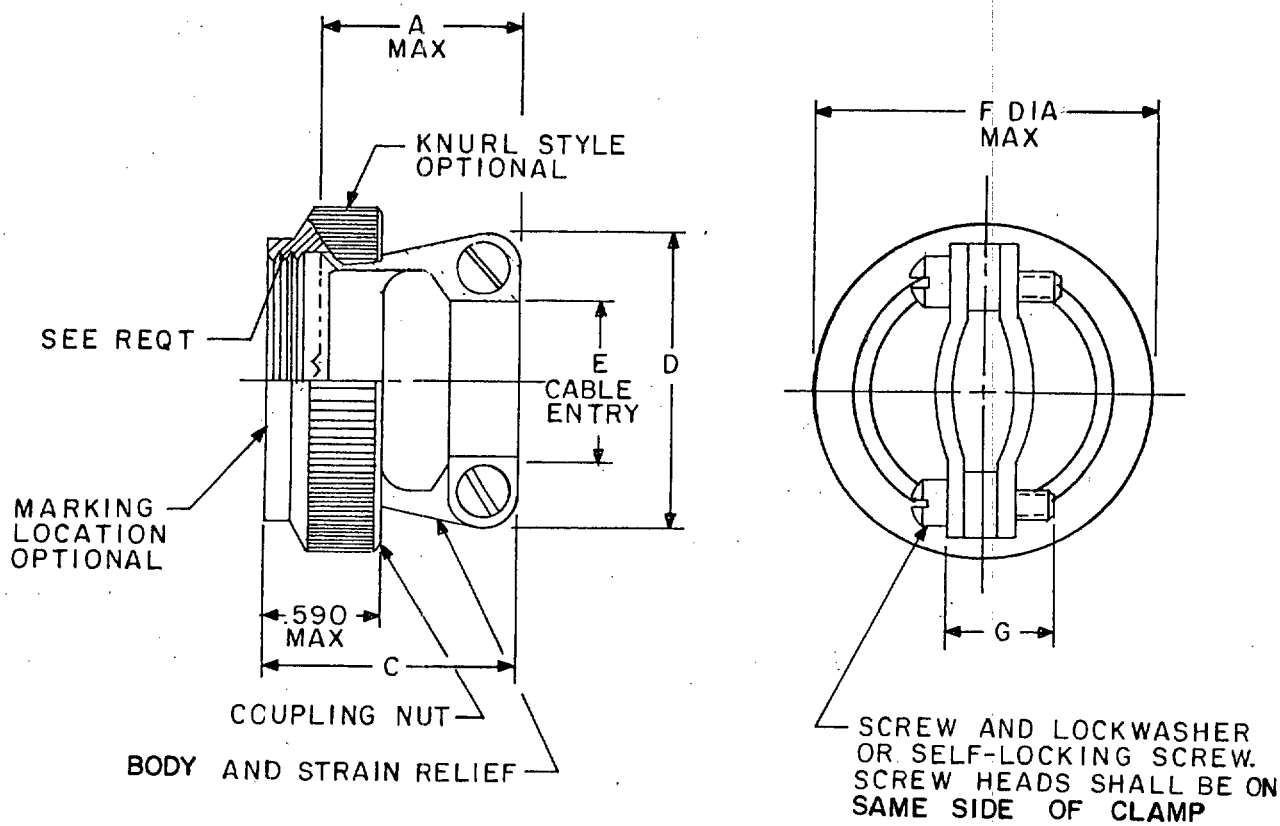


FIGURE 1. Dimensions and configurations (self-locking only).

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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Shell sizes and dimensions

Dash number	Connector Shell size		A max	C max	D max	E Cable entry		F max dia	G Screw Length ±.032	Screw size
	Series II	Series I				Max	Min			
8	8	9	.91 (23.1)	1.10 (27.9)	.85 (21.6)	.234 (5.94)	.098 (2.49)	.859 (21.82)	.500 (12.70)	6-32
10	10	11	.91 (23.1)	1.10 (27.9)	.90 (22.9)	.234 (5.94)	.153 (3.89)	.984 (24.99)	.500 (12.70)	6-32
12	12	13	1.01 (25.7)	1.20 (30.5)	1.10 (27.9)	.328 (8.33)	.190 (4.83)	1.156 (29.36)	.625 (15.88)	6-32
14	14	15	1.06 (26.9)	1.25 (31.8)	1.15 (29.2)	.457 (11.61)	.260 (6.60)	1.281 (32.54)	.750 (19.05)	6-32
16	16	17	1.16 (29.5)	1.35 (34.3)	1.30 (33.0)	.634 (16.10)	.283 (7.19)	1.406 (35.71)	.750 (19.05)	6-32
18	18	19	1.41 (35.9)	1.60 (40.6)	1.50 (38.1)	.614 (15.60)	.325 (8.26)	1.515 (38.51)	.750 (19.05)	8-32
20	20	21	1.51 (38.4)	1.70 (43.2)	1.60 (40.6)	.698 (17.73)	.343 (8.71)	1.641 (41.68)	.875 (22.23)	8-32
22	22	23	1.66 (42.2)	1.85 (47.0)	1.70 (43.2)	.823 (20.90)	.381 (9.68)	1.765 (44.86)	1.000 (25.40)	8-32
24	24	25	1.76 (44.7)	1.95 (49.5)	1.80 (45.7)	.853 (21.67)	.418 (10.62)	1.891 (48.03)	1.125 (28.58)	8-32

NOTES:
1. Dimensions are in inches.
2. Dimensions apply after plating.
3. Metric equivalents are in parentheses and are given for information only.

FIGURE 1. Dimensions and configurations (self-locking only) - Continued.

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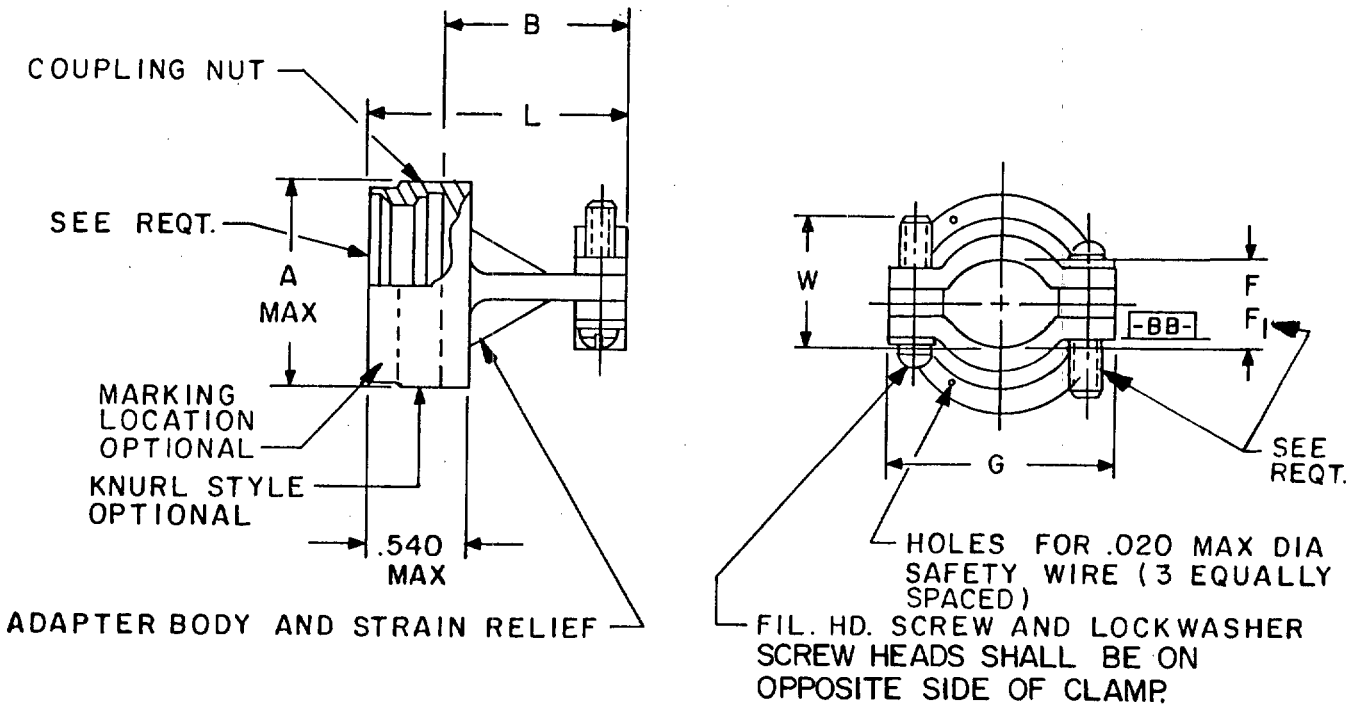


FIGURE 2. Dimensions and configurations (nonself-locking only).

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Shell sizes and dimensions

Dash number	Fits shell size		A max dia of coupling ring	B max	F max closed	F_1 Min. full open	G max	L max	W ±.032 screw length	Screw size
	Series II	Series I								
8	8	9	.75 (19.1)	.91 (23.1)	.098 (2.49)	.234 (5.94)	.85 (21.6)	1.10 (27.9)	.500 (12.70)	6-32
10	10	11	.85 (21.5)	.91 (23.1)	.153 (3.89)	.234 (5.94)	.90 (22.9)	1.10 (27.9)	.500 (12.70)	6-32
12	12	13	1.00 (25.4)	1.01 (25.7)	.190 (4.83)	.328 (8.33)	1.10 (27.9)	1.20 (30.5)	.625 (15.88)	6-32
14	14	15	1.10 (27.9)	1.06 (26.9)	.260 (6.60)	.457 (11.61)	1.15 (29.2)	1.25 (31.8)	.750 (19.05)	6-32
16	16	17	1.25 (31.8)	1.16 (29.5)	.283 (7.19)	.634 (16.10)	1.30 (33.0)	1.35 (34.3)	.750 (19.05)	6-32
18	18	19	1.40 (35.6)	1.41 (35.8)	.325 (8.26)	.614 (15.60)	1.50 (38.1)	1.60 (40.6)	.750 (19.05)	8-32
20	20	21	1.50 (38.1)	1.51 (38.4)	.343 (8.71)	.698 (17.73)	1.60 (40.6)	1.70 (43.2)	.875 (22.23)	8-32
22	22	23	1.65 (41.9)	1.66 (42.2)	.381 (9.68)	.823 (20.90)	1.70 (43.2)	1.85 (47.0)	1.000 (25.40)	8-32
24	24	25	1.75 (44.5)	1.76 (44.7)	.418 (10.62)	.853 (21.67)	1.80 (45.7)	1.95 (49.5)	1.125 (28.58)	8-32

NOTES:
1. Dimensions are in inches.
2. Dimensions apply after plating.
3. Metric equivalents are in parentheses and are given for information only.

FIGURE 2. Dimensions and configurations (nonself-locking only) - Continued.

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REQUIREMENTS:

Self-locking and nonself-locking styles:

Design and construction:

Dimensions and configurations: See figures 1 and 2.

Interface dimensions: In accordance with MIL-C-38999, series I and II.

Accessory: Consists of a coupling nut and body. The coupling nut shall be captivated to, and free to rotate on the body. This accessory will not accommodate connectors using size 8, 4, or 0 contacts. Figure 2 dimension F_1 is applicable when the end of screw is flush with -BB-.

Material and finish: See table I. Accessories shall be as shown in table I.

Clamp screws and lockwashers: 300 series corrosion-resistant steel, passivate.

TABLE I. Material and finish.

Figure	Material	Finish 1/
1	Aluminum alloy in accordance with MIL-C-85049	N and W
1	Corrosion-resistant steel, 300 series in accordance with QQ-S-763	S
2	Aluminum alloy in accordance with MIL-C-85049	A, N, and W

1/ A finish inactive for new design. N finish not for Navy use; Air Force use for space applications only. W is the prefered finish.

Self-locking styles only:

Self-locking device within the coupling nut shall be corrosion-resistant metal and shall provide a positive detent. Couplings with self-locking devices shall meet all the performance requirements specified herein for its specific category and the following additional tests.

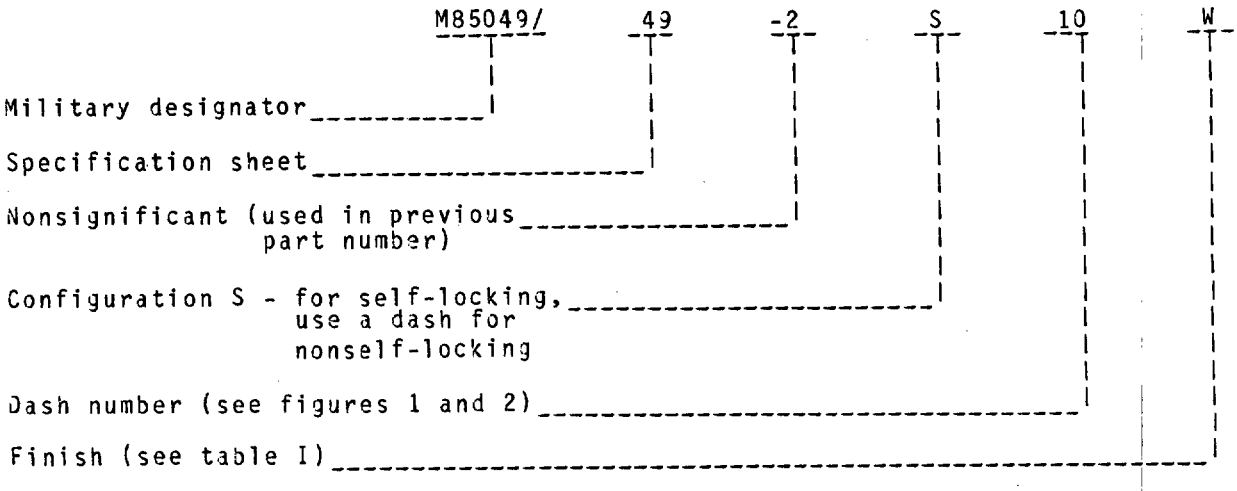
Life cycle: The coupling with the locking device engaged shall be rotated 60 full turns in a clockwise direction and 60 full turns counter clockwise, prior to standard qualification testing. Cycle speed shall not exceed one full turn per second.

Locking: Coupling nut torque shall be 80 percent of coupling thread strength values given for medium and light duty accessories as specified in table III of MIL-C-85049. In table III where N/A appears, the values shall be 80 percent of those values used in qualification testing for coupling thread strength. Tolerance shall be ±5 inch-pounds.

Vibration: Backshells shall be subjected to test parameters of applicable connector specifications. The coupling torque shall be within +20, -10 inch-pounds of the initial value.

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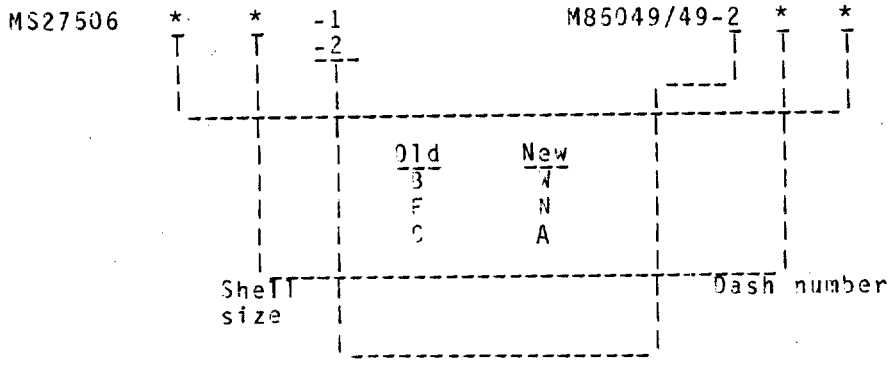
Part number: See example below:



Qualification: See MIL-C-85049, category 48. Accessory shall be tested with MIL-C-38999 series I connectors.

QPL evaluating activity: Naval Avionics Center, Code B/714, 6000 East 21st Street, Indianapolis, IN 46219-2189.

Supersession data: M85049/49-1XX is superseded by M85049/49-2XXX.



Parts qualified to MIL-C-85049/49A may be supplied for 1 year from the date of this document.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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